

INTERNATIONAL EXAMINATION CENTER
NETWORK DOCUMENTATION

Technical Network Documentation for the International Examination Center

Headquarters and Branch Enterprise Network

Field	Value
Project Type	Enterprise Networking, Security, and Automation
Primary Site	Headquarters / Main Branch — six-floor examination center
Secondary Site	Branch Office — five-floor examination center
Simulation Platform	Cisco Packet Tracer
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Course / Instructor	CCNA3 / Aisha al-Hadam
Document ID / Version	[DOC-NET-001] / [v1.0]
Classification	Academic Project — Internal

01. Project Overview

1.1 Business and Technical Context

The project is an international examination center supporting TOEFL, IELTS, Cisco, cybersecurity, and other international examinations. The operating environment requires a high density of computers in examination laboratories and stable connectivity throughout examination sessions to minimize service interruptions. Applicant data and examination results are sensitive; therefore, the design separates examination-user networks from administration and IT staff networks, server networks, the DMZ, and CCTV networks. The headquarters contains a central Data Center and a Control Unit monitoring room, while the secondary branch provides selected local services, including local DHCP and CCTV-related services, and relies on the headquarters for centralized network services.

1.2 Scope

The project scope covers the design and documentation of the headquarters and secondary-branch networks, WAN connectivity between the two sites through service providers, and wired and wireless infrastructure across the Access and Layer 3 Core/Distribution layers. It also covers the VLAN and IPv4/IPv6 addressing plans, interconnection addressing, DMZ, OSPFv2, NAT/PAT, ACLs, Site-to-Site IPsec VPN, QoS, device management and monitoring using NTP, Syslog, and SNMPv3, backup, and automation. The documentation also includes an inventory of devices, servers, cabling, patch-panel ports, inter-floor connectivity, and the physical and logical network designs.

1.3 Out of Scope

The project does not include purchasing or physically installing equipment in a real building, signing actual contracts with Internet service providers, or approving final purchase prices or production licenses without documented quotations. Packet Tracer is a simulation model rather than a complete production environment; therefore, simulation results do not replace real-world performance, security, or resilience testing. Production certificates, real identity management, legal compliance for applicant data, and integration with external examination platforms remain outside the academic model unless otherwise required by the instructor.

1.4 Design Objectives

The design aims to provide a scalable hierarchical enterprise network for the examination center, with clear separation between examination users, administration, employees/IT, servers, CCTV, printers/IoT, voice, guests, and management traffic. It also provides secure HQ-to-Branch connectivity, WAN resilience, Layer 3 routing, ACL, NAT/PAT, DMZ, IPsec VPN, QoS, centralized services, and network monitoring. The addressing and VLAN plan is standardized by site so that the logical topology, physical design, endpoint tables, and configuration records remain consistent.

02. Site and Floor Allocation

Site / Floor	Functional Area	Network Role	Observed / Planned Equipment
HQ / Ground Floor	Central Data Center, Control Unit, Monitoring	Core services, security, routing, central services	2× L3 Core/Distribution Switches, 2× Routers, Control Unit / Access Switches, Servers, 2× Firewalls, UPS
HQ / Floor 1	Reception	Reception access and administrative services	5 PCs, 5 Cameras, 5 Printers, Floor Rack / Access Switch
HQ / Floor 2	Examination Laboratories and Waiting Areas	Exam-user access	54 Exam PCs, 60 Cameras, 6 APs, Access Switches / Rack
HQ / Floor 3	Examination Laboratories and Waiting	Exam-user access	54 Exam PCs, 60 Cameras, 6 APs, Access Switches / Rack

Site / Floor	Functional Area	Network Role	Observed / Planned Equipment
	Areas		
HQ / Floor 4	Examination Laboratories and Waiting Areas	Exam-user access	54 Exam PCs, 60 Cameras, 6 APs, Access Switches / Rack
HQ / Floor 5	Administration and IT Support	IT/Admin and Management Access	16 IT/Employee PCs, 3 Administration PCs, 10 Cameras, 3 Printers, Access Switch / Rack
Branch 2 / Ground Floor	Control / Reception Area	Branch access and local services	5 PCs, 8 Cameras, 1 Printer, Access Switches / Rack
Branch 2 / Floor 1	Reception	Reception access and administrative services	5 PCs, 5 Cameras, 5 Printers, Access Switch / Rack
Branch 2 / Floor 2	Examination Laboratories and Waiting Areas	Exam-user access	54 Exam PCs, 60 Cameras, 6 APs, Access Switches / Rack
Branch 2 / Floor 3	Examination Laboratories and Waiting Areas	Exam-user access	54 Exam PCs, 60 Cameras, 6 APs, Access Switches / Rack
Branch 2 / Floor 4	Administration and IT Support	IT/Admin and Management Access	16 IT/Employee PCs, 3 Administration PCs, 10 Cameras, 3 Printers, Access Switch / Rack

03. Network Architecture

Block	Location	Function	Primary Links	Security Boundary
Access Layer	Each floor rack	Endpoint / Access Layer	Patch Panels, Access Switches, Cat6 endpoint connections	User / Device VLANs
Distribution / Core	HQ and Branch Ground Floor network racks	Distribution / Core Layer	Two Layer 3 Multilayer Switches per site; inter-VLAN routing and aggregation	Routing Boundary
WAN Edge	HQ and Branch router racks	ISP Connectivity, NAT, VPN	Router-to-ISP and Router-to-Firewall links	Untrusted WAN
DMZ	HQ server / security block	DMZ services including Web and related infrastructure services	Firewall-controlled segment	Internet-facing services
Management	HQ Floor 5 / IT and central services	SSH, SNMPv3, NTP, Backups	Management network connections	Restricted Management VLAN
Security / DMZ	HQ and Branch firewalls	Security boundary; HQ firewall	Redundant routed links	Security / WAN Boundary

Block	Location	Function	Primary Links	Security Boundary
		connects to DMZ and internal security boundary		
WAN / Edge	HQ and Branch	External WAN and Internet connectivity	Two routers per site; two ISP paths	Security / WAN Boundary

04. Physical Network Design

HQ

Rack ID	Site / Floor	Rack Function	Size	Equipment / Devices	Power	Cabling
R-HQ-GF-01	HQ Ground Floor	Routers Rack	24U	HQ-R1; HQ-R2	UPS	Fiber optic
R-HQ-GF-02	HQ Ground Floor	Layer 3 Switches Rack	24U	HQ-CORE1; HQ-CORE2	UPS	Fiber optic
R-HQ-GF-03	HQ Ground Floor	Control Unit Switch Rack	12U	HQ-GF-SW1; HQ-GF-SW2	UPS	Cat6 UTP + Fiber uplinks
R-HQ-GF-04	HQ Ground Floor	Servers Rack	24U	Server0; Server1; Server2; Server3; Camera_Server	UPS	Cat6 UTP + Fiber uplinks
R-HQ-GF-05	HQ Ground Floor	Firewalls Rack	12U	ASA1; ASA3	UPS	Fiber / Copper
R-HQ-GF-06	HQ Ground Floor	UPS Rack	24U	UPS units	Utility + Battery	Power only
R-HQ-F5-01	HQ Floor 5	Management / Floor Rack	12U	HQ-F5-SW1	UPS	Cat6 + Fiber uplinks

Branch 2

Rack ID	Site / Floor	Rack Function	Size	Equipment / Devices	Power	Cabling
R-BR-GF-01	Branch 2 Ground Floor	Network Core / Router Rack	24U	BR-R1; BR-R2; BR-CORE1; BR-CORE2	UPS	Fiber optic
R-BR-GF-02	Branch 2 Ground Floor	Access / Server Rack	24U	BR-GF-SW1; BR-GF-SW2; Branch_DHCP_Server; Branch_Camera_Server; ASA22	UPS	Cat6 + Fiber
R-BR-F1-01	Branch 2 Floor 1	Floor Access Rack	12U	BR-F1-SW1	UPS	Cat6 + Fiber uplinks
R-BR-F2-01	Branch 2 Floor 2	Floor Access Rack	24U	BR-F2-SW1; BR-F2-SW2; BR-F2-SW3	UPS	Cat6 + Fiber uplinks
R-BR-F3-01	Branch 2 Floor 3	Floor Access Rack	24U	BR-F3-SW1; BR-F3-SW2; BR-F3-SW3	UPS	Cat6 + Fiber uplinks
R-BR-F4-	Branch 2	Floor Access	12U	BR-F4-SW1	UPS	Cat6 + Fiber

Rack ID	Site / Floor	Rack Function	Size	Equipment / Devices	Power	Cabling
01	Floor 4	Rack				uplinks

Figure A1 — Headquarters Ground Floor Physical Design

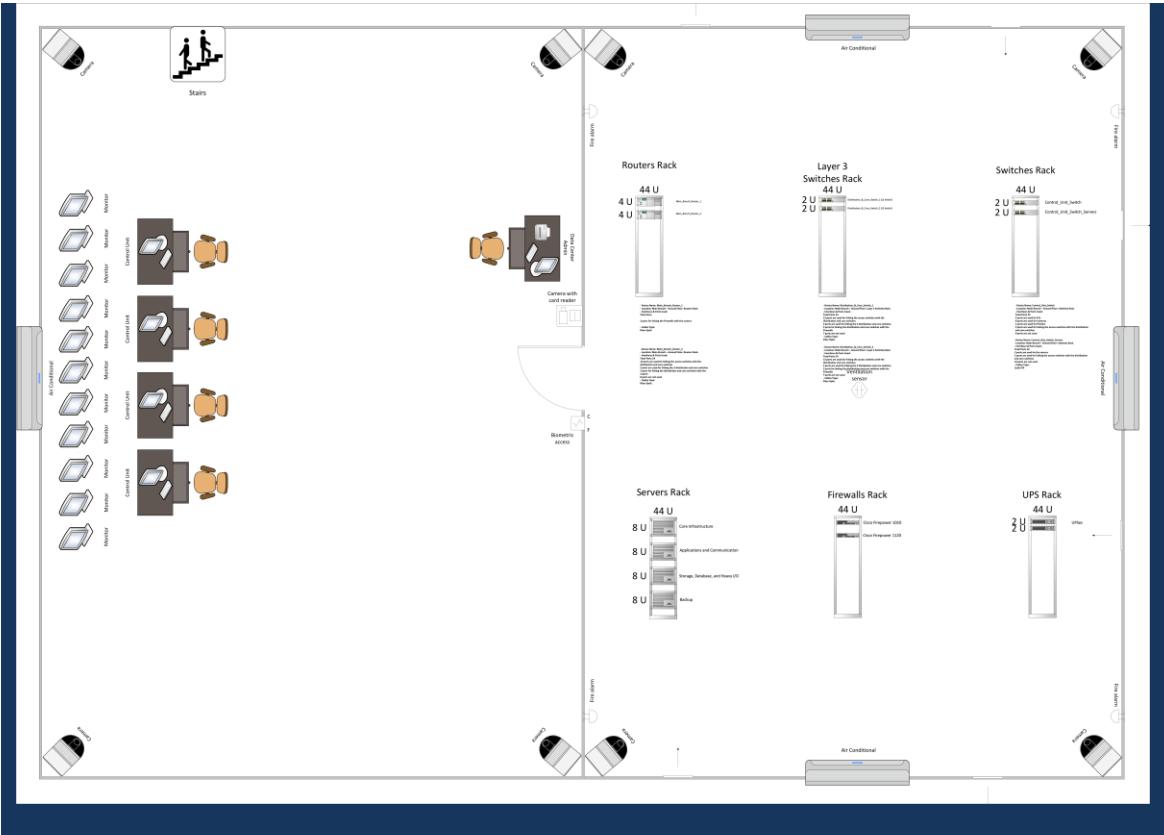


Figure A2 — Headquarters Floor 1 Physical Design



Figure A3 — Headquarters Floor 2 Physical Design

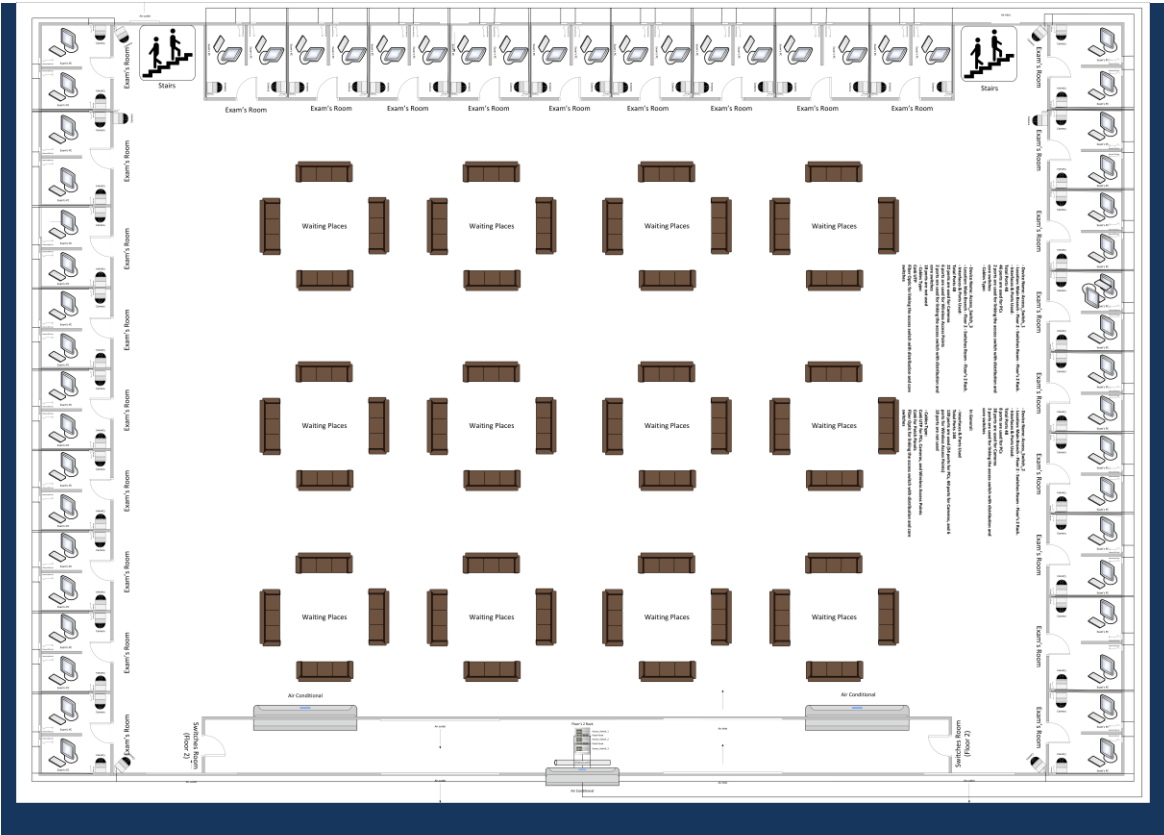


Figure A4 — Headquarters Floor 3 Physical Design

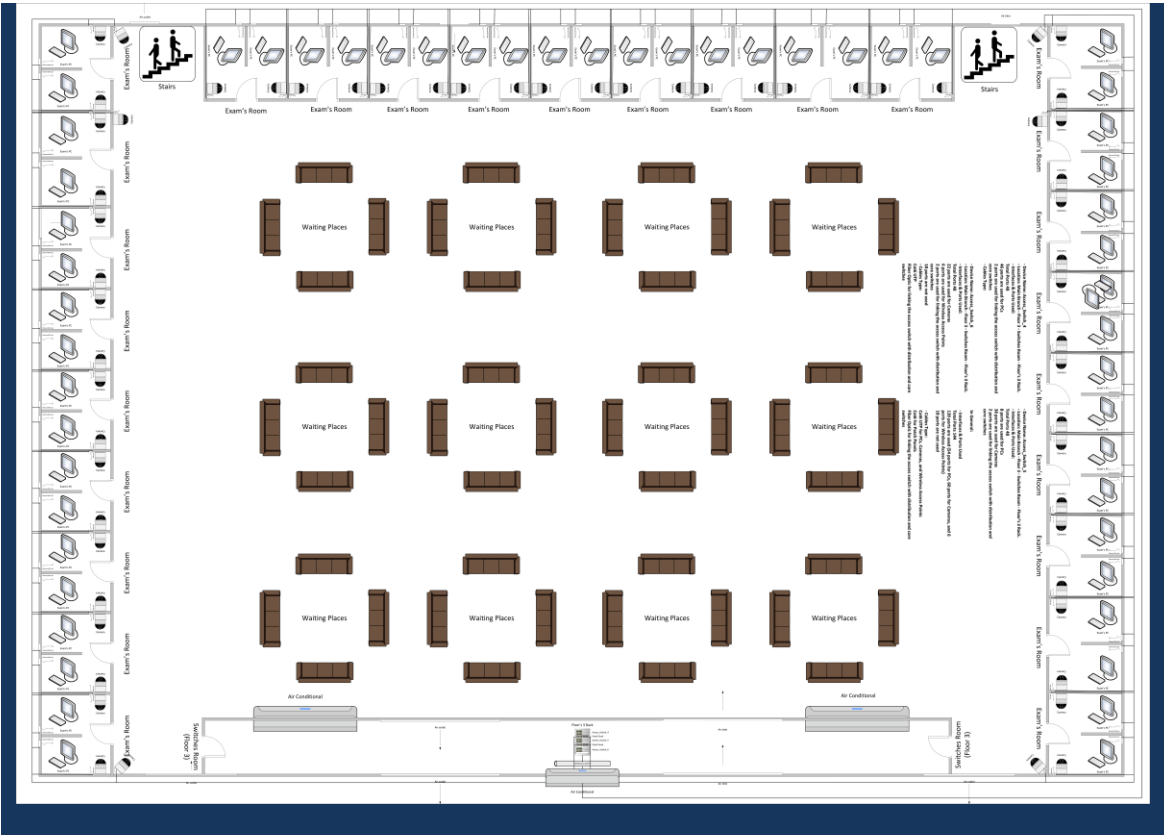


Figure A5 — Headquarters Floor 4 Physical Design

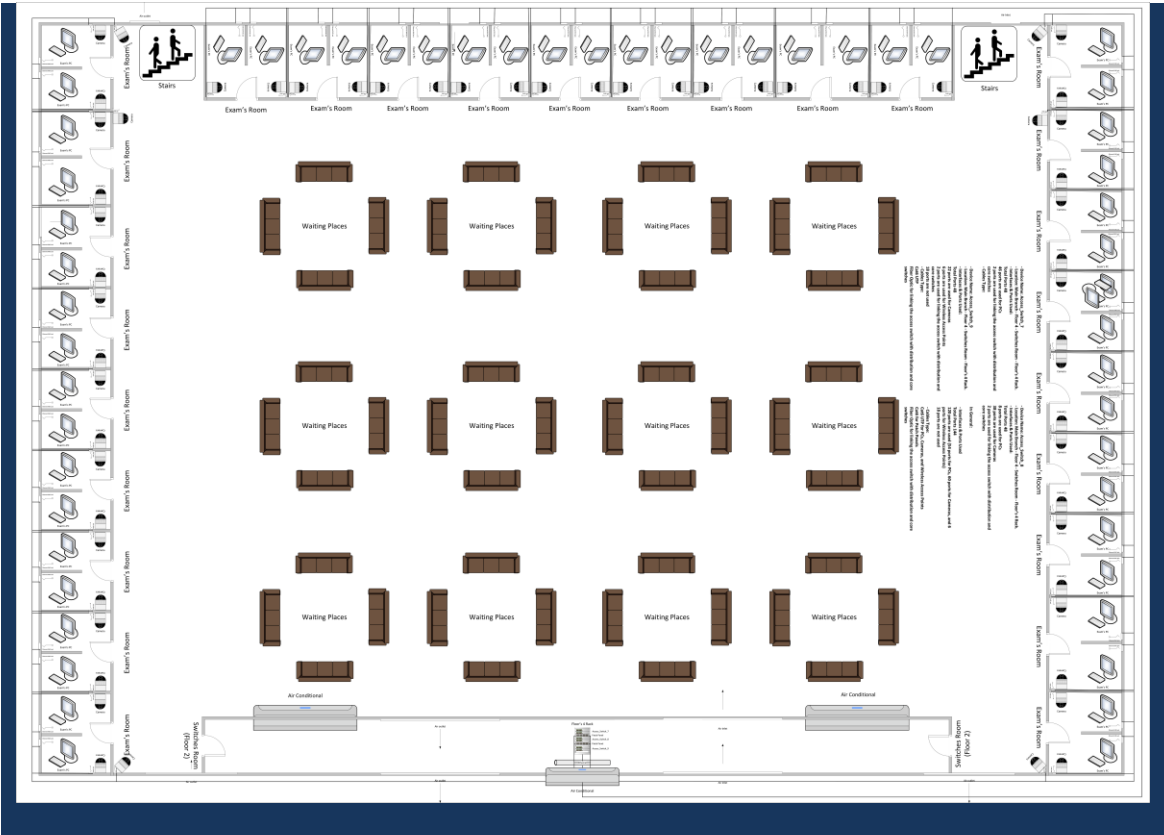


Figure A6 — Headquarters Floor 5 Physical Design

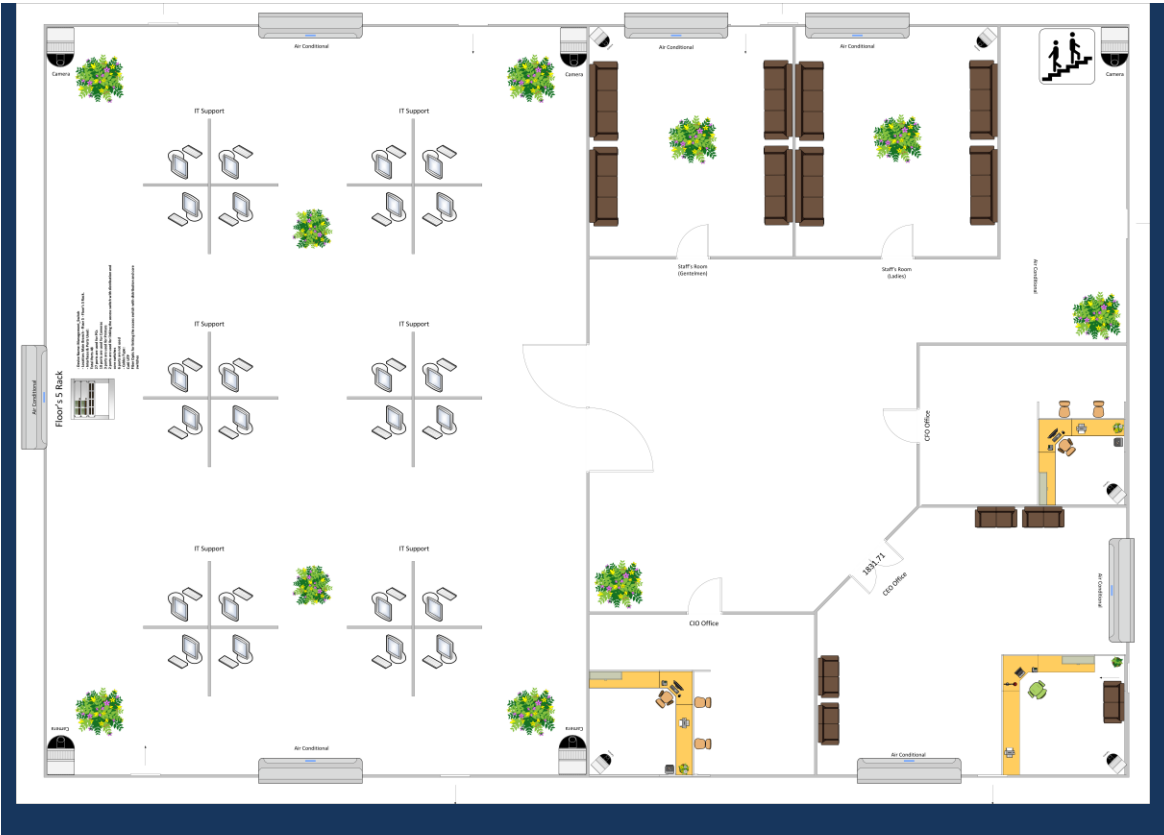


Figure A7 — Secondary Branch Physical Design

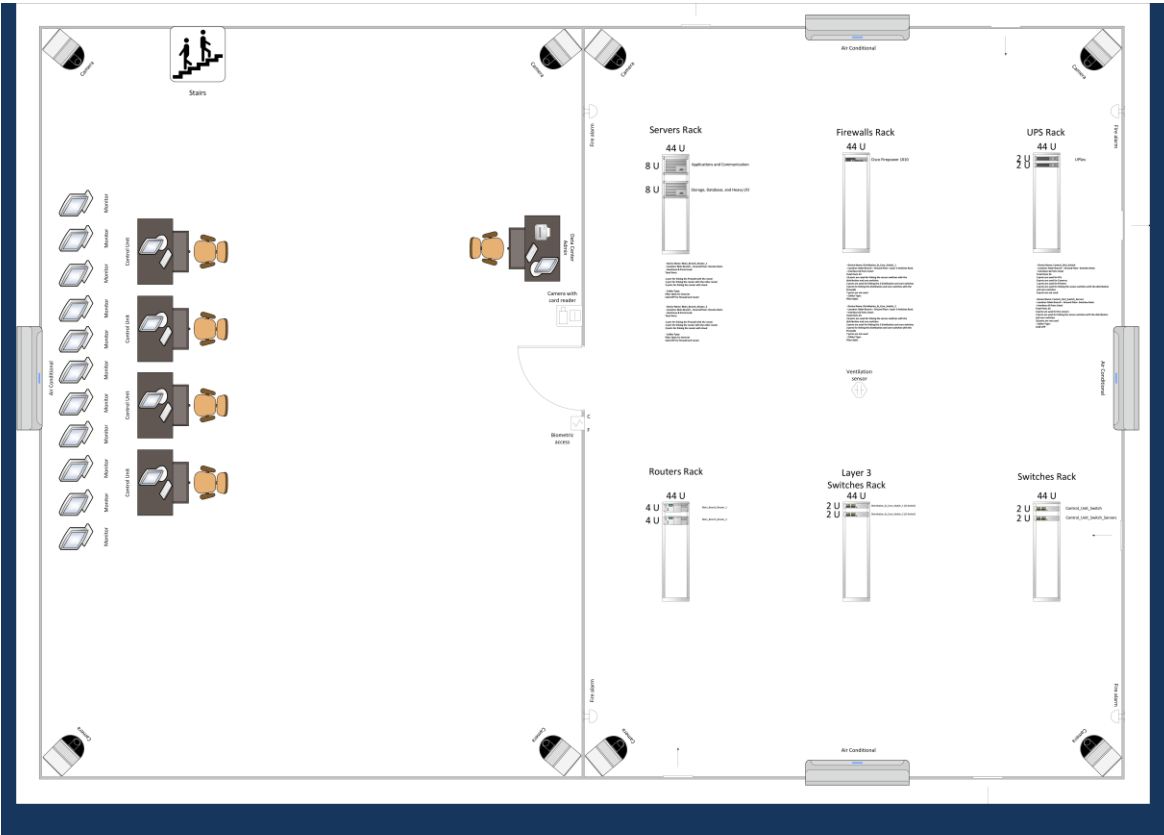


Figure A8 — Secondary Branch Physical Design



Figure A9 — Secondary Branch Physical Design

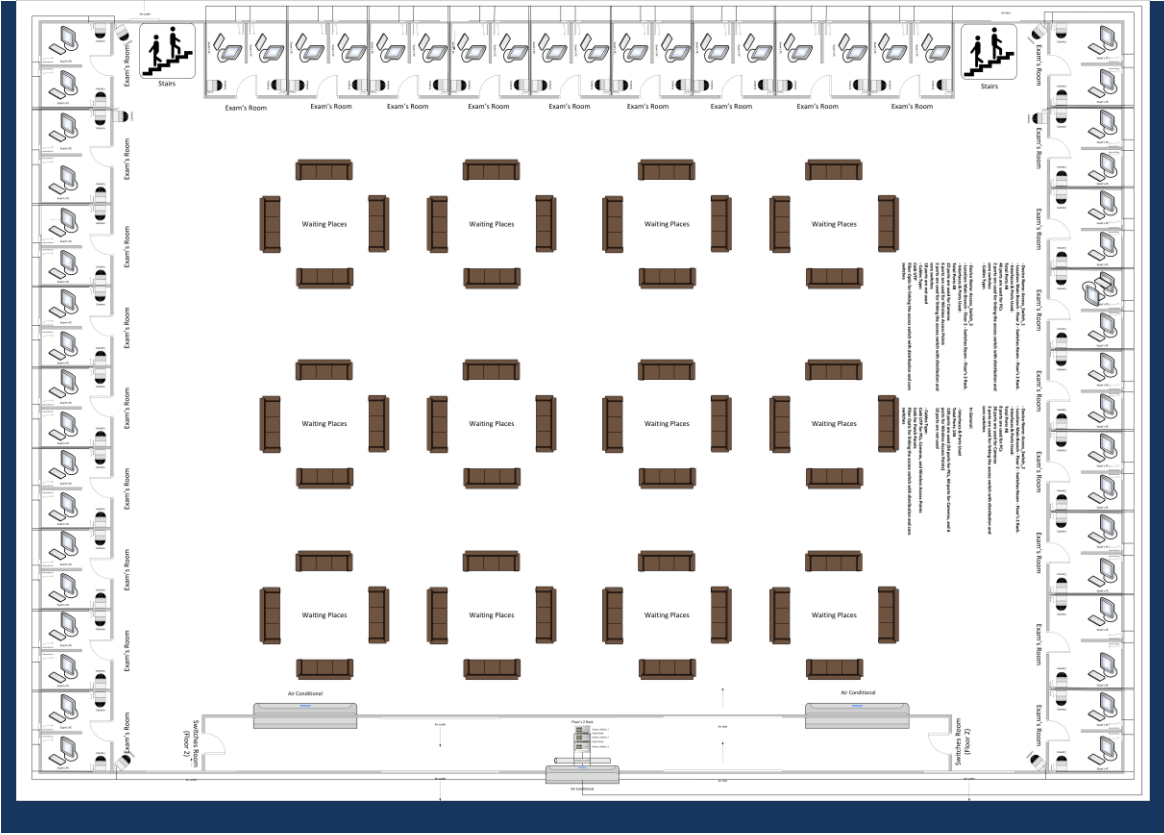


Figure A10 — Secondary Branch Physical Design

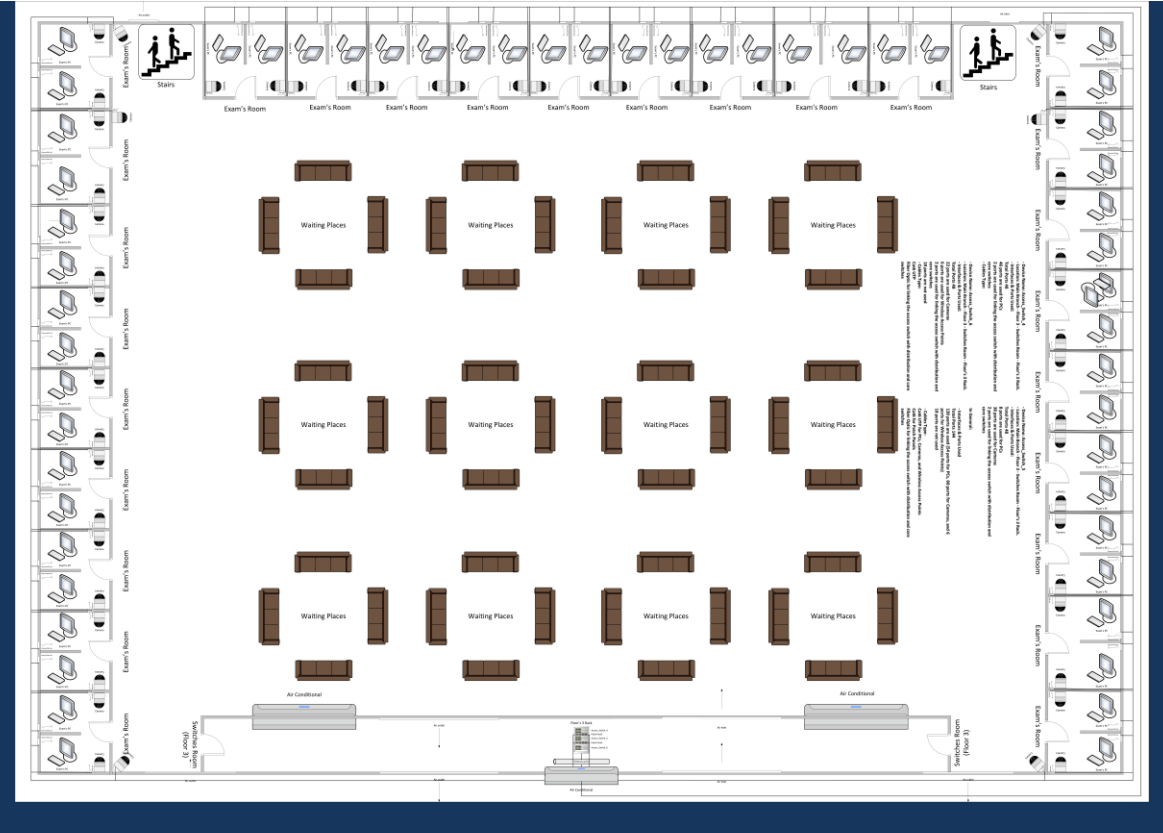
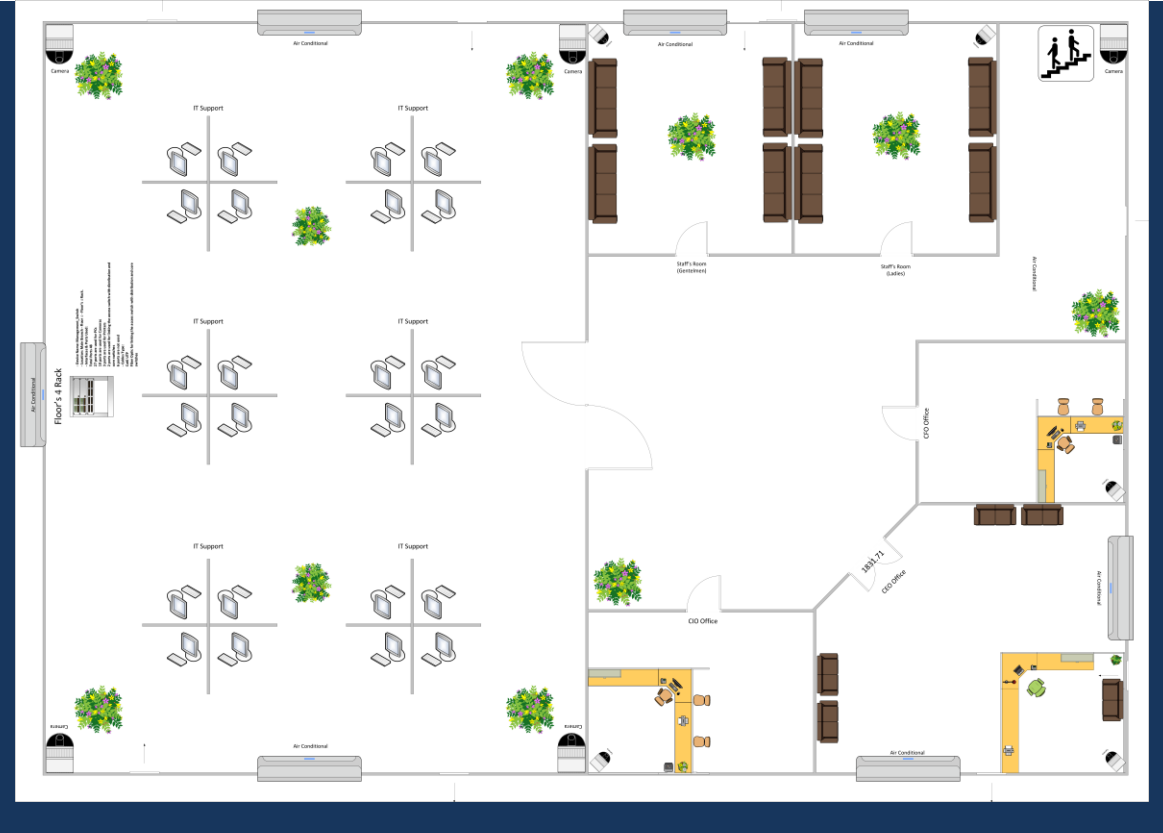


Figure A7 — Secondary Branch Physical Design



05. Logical Network Design

Figure 1 — Packet Tracer Logical Topology

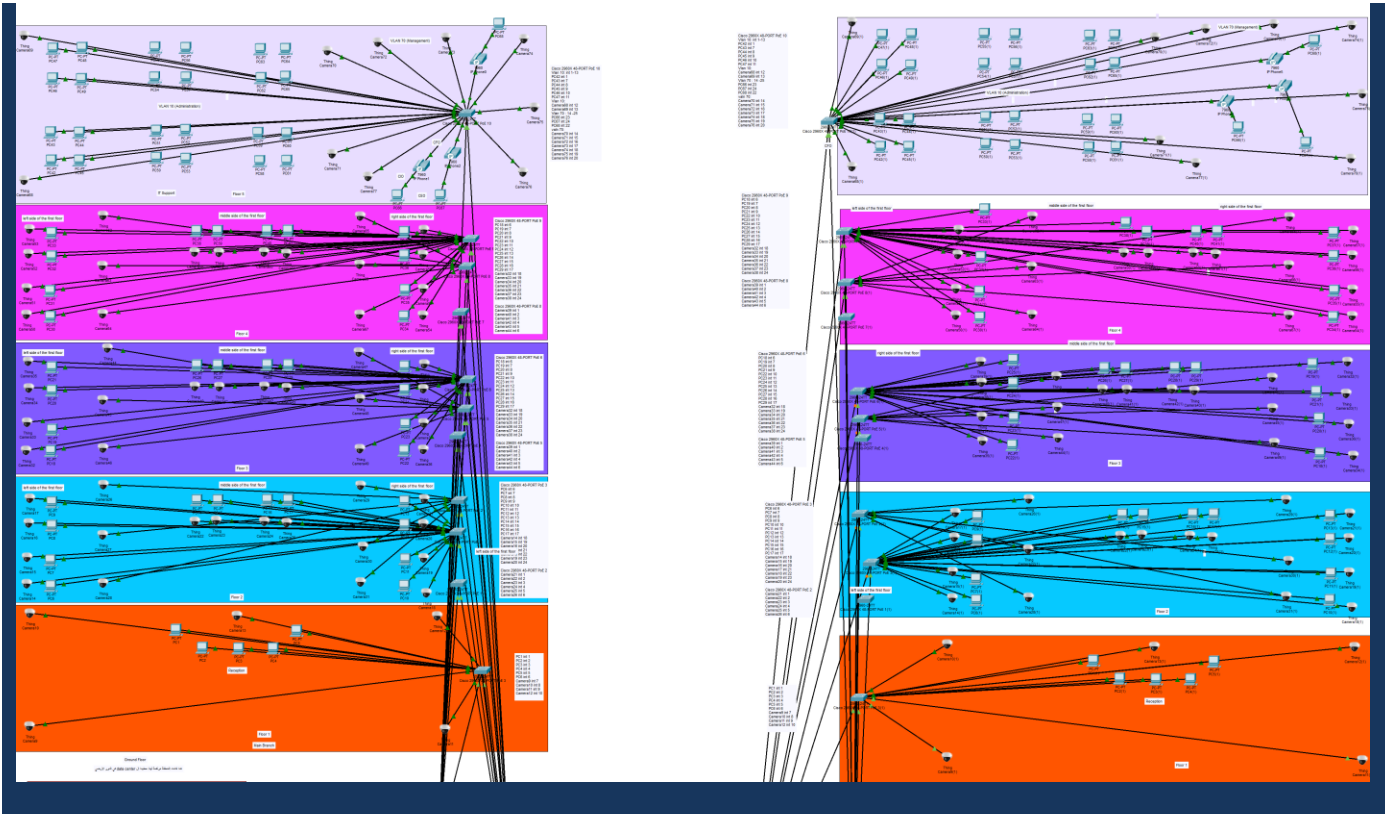
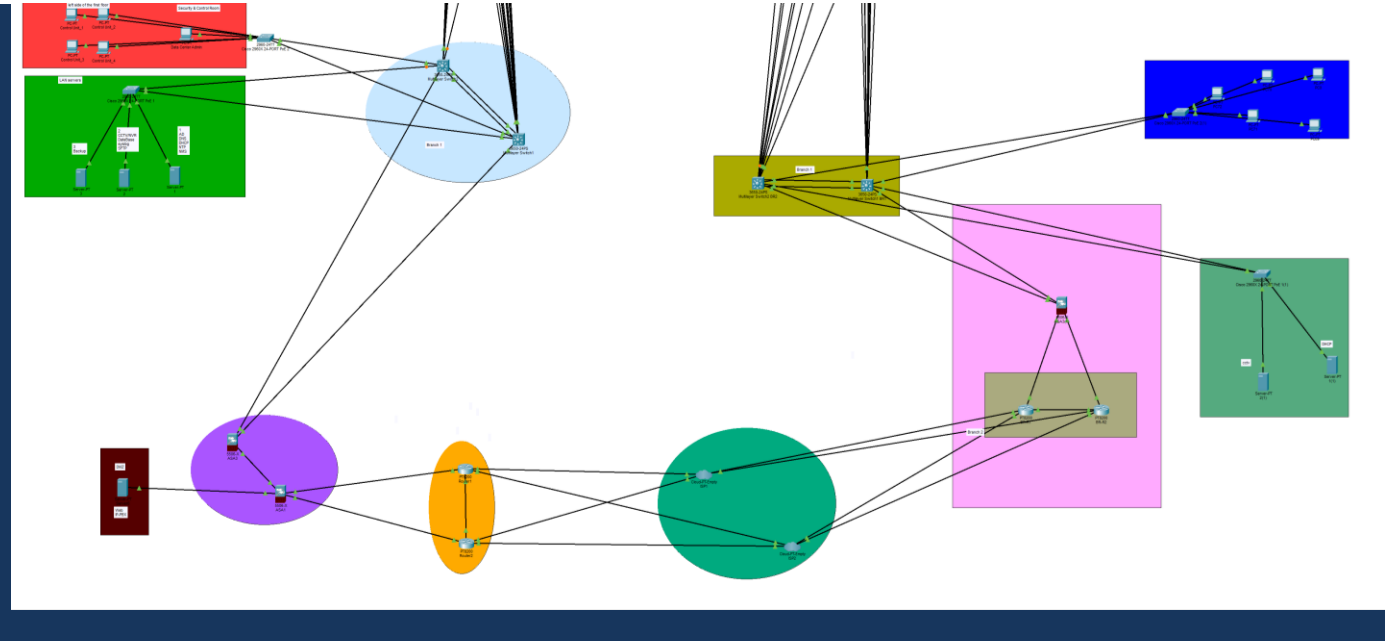


Figure 2— Packet Tracer Logical Topology



06. VLAN and IP Addressing Plan

VLAN ID	Name	Purpose	Site/Floor	Subnet	Gateway	DHCP	ACL / Zone
10	ADMIN	Administration / control-unit and IT admin PCs	HQ + Branch	HQ 10.10.10.0/24; Branch 10.20.10.0/24	HQ .1 / Branch .1	Yes — local site DHCP	Admin policy
20	EMPLOYEES	Reception / employee endpoints	HQ + Branch	HQ 10.10.20.0/24; Branch 10.20.20.0/24	HQ .1 / Branch .1	Yes — local site DHCP	Employee policy
30	TEST_CENTER	Examination laboratories	HQ + Branch	HQ 10.10.30.0/24; Branch 10.20.30.0/24	HQ .1 / Branch .1	Yes — local site DHCP	Exam policy
40	CCTV	IP cameras / CCTV	HQ + Branch	HQ 10.10.40.0/24; Branch 10.20.40.0/24	HQ .1 / Branch .1	Yes — local site DHCP / reservations	CCTV only
50	VOICE	Voice services / IP-PBX	HQ + Branch	HQ 10.10.50.0/24; Branch 10.20.50.0/24	HQ .1 / Branch .1	Yes — local site DHCP	QoS / voice
60	SERVERS	Infrastructure and application servers	HQ + Branch	HQ 10.10.60.0/24; Branch 10.20.60.0/24	HQ .1 / Branch .1	No — static	Server policy
70	MANAGEMENT	Network management	HQ + Branch	HQ 10.10.70.0/24; Branch 10.20.70.0/24	HQ .1 / Branch .1	No — static	Admin-only
80	GUEST	Guest access	HQ + Branch	HQ 10.10.80.0/24; Branch 10.20.80.0/24	HQ .1 / Branch .1	Yes — local site DHCP	Internet only
90	PRINTERS_IOT	Printers and IoT endpoints	HQ + Branch	HQ 10.10.90.0/24; Branch 10.20.90.0/24	HQ .1 / Branch .1	Yes — local site DHCP / reservations	Printer/IoT policy
100	DMZ	DMZ services	HQ only	10.10.100.0/24	10.10.100.1	No — static	Firewall controlled
999	NATIVE	Native VLAN; no user endpoints	HQ + Branch	No user subnet	N/A	N/A	Unused/native only

07. Device and Service Inventory

Asset ID	Hostname	Role	Site/Floor	Interfaces / Ports
R-HQ-01	HQ-R1	HQ Border / WAN Router	HQ Ground Floor	Gi0/0/0 → Router 2Gi0/0/1 → Firewall 1Gi0/0/2 → ISP 1Gi0/0/3 → ISP 2
R-HQ-02	HQ-R2	HQ Border / WAN Router	HQ Ground Floor	Gi0/0/0 → Router 1Gi0/0/1 → Firewall 1Gi0/0/2 → ISP 1Gi0/0/3 → ISP 2
L3-HQ-01	HQ-CORE1	Layer 3 Distribution / Core	HQ Ground Floor	Port-Channel1 → L3-HQ-02Gi1/0/2 → Firewall 3Access uplinks → Floor Switches
L3-HQ-02	HQ-CORE2	Layer 3 Distribution / Core	HQ Ground Floor	Port-Channel1 → L3-HQ-01Gi1/0/2 → Firewall 3Access uplinks → Floor Switches
ACC-HQ-GF-01	HQ-GF-SW1	Control Unit Access Switch	HQ Ground Floor	5 PC Ports8 CCTV Ports1 Printer / IoT Port2 Uplinks → Core
ACC-HQ-F5-01	HQ-F5-SW1	IT & Administration Access Switch	HQ Floor 5	19 PC Ports10 CCTV Ports3 Printer Ports2 Uplinks14 Spare Ports
SRV-HQ-01	Core_Infrastructure_Server	AD DS, DNS, DHCP, NTP & NMS/Zabbix	HQ Ground Floor — Data Center	Server NIC → Servers VLANOptional Management Access → VLAN 70
R-BR-01	BR-R1	Branch Border / WAN Router	Branch Ground Floor	→ Branch Firewall; WAN → ISP 1 / ISP 2
R-BR-02	BR-R2	Branch Border / WAN Router	Branch Ground Floor	→ Branch Firewall; WAN → ISP 1 / ISP 2
L3-BR-01	Branch_Multilayer_Switch_1	Branch Layer 3 Distribution/Core	Branch Ground Floor	Port-Channel → L3-BR-02; firewall uplink; Access Switch uplinks
L3-BR-02	Branch_Multilayer_Switch_2	Branch Layer 3 Distribution/Core	Branch Ground Floor	Port-Channel → L3-BR-01; firewall uplink; Access Switch uplinks

7.1 Detailed Port Utilization from Visio

HQ

Device	Location	Total Ports	Used Ports	Unused Ports	Port Allocation	Cable Type
HQ-GF-SW1	HQ Ground Floor / Switches Rack	24	16	8	5 PCs; 8 Cameras; 1 Printer; 2 Uplinks	Cat6 UTP
HQ-GF-SW1_Servers	HQ Ground Floor / Switches Rack	24	6	18	4 Servers; 2 Uplinks	Cat6 UTP

Device	Location	Total Ports	Used Ports	Unused Ports	Port Allocation	Cable Type
HQ-F1-SW1	HQ Floor 1 / Floor 1 Rack	24	17	7	5 PCs; 5 Cameras; 5 Printers; 2 Uplinks	Cat6 UTP + Fiber
HQ-F2/F3/F4-SW1 ×3	Floors 2–4 / Floor Racks	48	48	0	46 PCs; 2 Uplinks per switch	Cat6 UTP + Fiber
HQ-F2/F3/F4-SW2 ×3	Floors 2–4 / Floor Racks	48	48	0	8 PCs; 38 Cameras; 2 Uplinks per switch	Cat6 UTP + Fiber
HQ-F2/F3/F4-SW3 ×3	Floors 2–4 / Floor Racks	48	30	18	22 Cameras; 6 APs; 2 Uplinks per switch	Cat6 UTP + Fiber
HQ-F5-SW1	HQ Floor 5 / Floor 5 Rack	48	34	14	19 PCs; 10 Cameras; 3 Printers; 2 Uplinks	Cat6 UTP + Fiber
HQ-CORE1	HQ Ground Floor / Layer 3 Switches Rack	24	17	7	13 Access Links; 2 Inter-Core; 2 Firewall Links	Fiber Optic
HQ-CORE2	HQ Ground Floor / Layer 3 Switches Rack	24	17	7	13 Access Links; 2 Inter-Core; 2 Firewall Links	Fiber Optic
HQ-R1	HQ Ground Floor / Routers Rack	4	4	0	Firewall; Router; Internet	Fiber Optic / Cat6
HQ-R2	HQ Ground Floor / Routers Rack	4	4	0	Firewall; Router; Internet	Fiber Optic / Cat6

Branch 2

Device	Location	Total Ports	Used Ports	Unused Ports	Port Allocation	Cable Type
BR-GF-SW1	Branch Ground Floor / Switches Rack	24	16	8	5 PCs; 8 Cameras; 1 Printer; 2 Uplinks	Cat6 UTP
BR-GF-SW1_Servers	Branch Ground Floor / Switches Rack	24	6	18	4 Servers; 2 Uplinks	Cat6 UTP
BR-F1-SW1	Branch Floor 1 / Floor 1 Rack	24	17	7	5 PCs; 5 Cameras; 5 Printers; 2 Uplinks	Cat6 UTP + Fiber
BR-F2/F3-SW1 ×2	Floors 2–3 / Floor Racks	48	48	0	46 PCs; 2 Uplinks per switch	Cat6 UTP + Fiber
BR-F2/F3-SW2 ×2	Floors 2–3 / Floor Racks	48	48	0	8 PCs; 38 Cameras; 2 Uplinks per switch	Cat6 UTP + Fiber
BR-F2/F3-SW3 ×2	Floors 2–3 / Floor Racks	48	30	18	22 Cameras; 6 APs; 2 Uplinks per switch	Cat6 UTP + Fiber
BR-F4-SW1	Branch Floor 4 / Floor 4 Rack	48	34	14	19 PCs; 10 Cameras; 3 Printers; 2 Uplinks	Cat6 UTP + Fiber
BR-CORE1	Branch Ground Floor / Layer 3 Switches Rack	24	17	7	10 Access Links; 2 Inter-Core; 2 ASA22 Links; 2 Server Links	Fiber Optic
BR-CORE2	Branch Ground Floor / Layer 3 Switches Rack	24	17	7	10 Access Links; 2 Inter-Core; 2 ASA22 Links; 2 Server Links	Fiber Optic

Device	Location	Total Ports	Used Ports	Unused Ports	Port Allocation	Cable Type
BR-R1	Branch Ground Floor / Routers Rack	4	4	0	ASA22; Router; ISP1; ISP2	Fiber Optic / Cat6
BR-R2	Branch Ground Floor / Routers Rack	4	4	0	ASA22; Router; ISP1; ISP2	Fiber Optic / Cat6

7.2 Router Device Documentation

Device	Model	Description	Location / Site	IOS / Image	License
HQ-R1	Cisco Catalyst C8200-1N-4T	HQ WAN / Edge Router 1	HQ Ground Floor — Routers Rack	Cisco IOS XE 17.x	Ipbasek9
HQ-R2	Cisco Catalyst C8200-1N-4T	HQ WAN / Edge Router 2	HQ Ground Floor — Routers Rack	Cisco IOS XE 17.x	Ipbasek9
BR-R1	Cisco Catalyst C8200-1N-4T	Branch WAN / Edge Router 1	Secondary Branch	Cisco IOS XE 17.x	Ipbasek9
BR-R2	Cisco Catalyst C8200-1N-4T	Branch WAN / Edge Router 2	Secondary Branch	Cisco IOS XE 17.x	Ipbasek9

HQ-R1

Interface	Description	IPv4 Address	IPv6 Address	Routing
Gi0/0/0	Inter-Router Link → Router 2	10.10.251.1/30	2001:db8:acad:251::1/64	OSPFv2
Gi0/0/1	Link → ASA1	10.10.252.2/30	2001:db8:acad:252::2/64	OSPFv2 + IPsec VPN
Gi0/0/2	WAN Link → ISP 1	10.10.250.2/30	N/A	Default Route / ISP 1
Gi0/0/3	WAN Link → ISP 2	10.10.250.10/30	N/A	Default Route / ISP 2

HQ-R2

Interface	Description	IPv4 Address	IPv6 Address	Routing
Gi0/0/0	Inter-Router Link → Router 1	10.10.251.2/30	2001:db8:acad:251::2/64	OSPFv2
Gi0/0/1	Link → ASA1	10.10.252.6/30	2001:db8:acad:252:1::2/64	OSPFv2 + IPsec VPN
Gi0/0/2	WAN Link → ISP 1	10.10.250.6/30	N/A	Default Route /

Interface	Description	IPv4 Address	IPv6 Address	Routing
				ISP 1
Gi0/0/3	WAN Link → ISP 2	10.10.250.14/30	N/A	Default Route / ISP 2

BR-R1

Interface	Description	IPv4 Address	IPv6 Address	Routing
Gi0/0/0	Inter-Router Link → BR-R2	10.20.251.1/30	2001:db8:acad:251:10::1/64	OSPFv2
Gi0/0/1	Link → ASA22	10.20.252.2/30	2001:db8:acad:252:10::2/64	OSPFv2 + IPsec VPN
Gi0/0/2	WAN Link → ISP 1	10.20.250.2/30	N/A	Default Route / ISP 1
Gi0/0/3	WAN Link → ISP 2	10.20.250.10/30	N/A	Default Route / ISP 2

BR-R2

Interface	Description	IPv4 Address	IPv6 Address	Routing
Gi0/0/0	Inter-Router Link → BR-R1	10.20.251.2/30	2001:db8:acad:251:10::2/64	OSPFv2
Gi0/0/1	Link → ASA22	10.20.252.6/30	2001:db8:acad:252:10::6/64	OSPFv2 + IPsec VPN
Gi0/0/2	WAN Link → ISP 1	10.20.250.6/30	N/A	Default Route / ISP 1
Gi0/0/3	WAN Link → ISP 2	10.20.250.14/30	N/A	Default Route / ISP 2

7.3 LAN Switch Device Documentation**HQ-Main Branch**

Device	Model	Description	Management IP	IOS / Image	VTP Domain / Mode
HQ-CORE1	Cisco Catalyst C9300-24T-A	Test Center Core / Distribution Layer 3 — Primary	10.10.70.2/24	Cisco IOS XE 17.x	TEST-CENTER / Transparent
HQ-CORE2	Cisco Catalyst C9300-24T-A	Test Center Core / Distribution Layer 3 — Secondary	10.10.70.3/24	Cisco IOS XE 17.x	TEST-CENTER / Transparent
HQ-GF-SW1	Cisco Catalyst WS-C2960- 24PC-L	Ground Floor Control Unit Access Switch	10.10.70.11/24	Cisco IOS 15.x	TEST-CENTER / Transparent
HQ-GF-SW2	Cisco Catalyst WS-C2960- 24PC-L	Ground Floor Data Center / Server Access Switch	10.10.70.12/24	Cisco IOS 15.x	TEST-CENTER / Transparent

Device	Model	Description	Management IP	IOS / Image	VTP Domain / Mode
HQ-F1-SW1	Cisco Catalyst WS-C2960-24PC-L	Floor 1 Reception Access Switch	10.10.70.13/24	Cisco IOS 15.x	TEST-CENTER / Transparent
HQ-F2/F3/F4-SW1–SW3 (9)	Cisco Catalyst WS-C2960-48PST-L	Floors 2–4 Examination, CCTV and AP Access Switches — 3 per floor	10.10.70.14–10.10.70.22/24	Cisco IOS 15.x	TEST-CENTER / Transparent
HQ-F5-SW1	Cisco Catalyst WS-C2960-48PST-L	Floor 5 IT / Administration Access Switch	10.10.70.22/24	Cisco IOS 15.x	TEST-CENTER / Transparent

Branch 2

Device	Model	Description	Management IP	IOS / Image	VTP Domain / Mode
BR-CORE1	Cisco Catalyst C9300-24T-A	Branch Core / Distribution Layer 3 — Primary	10.20.70.2/24	Cisco IOS XE 17.x	TEST-CENTER / Transparent
BR-CORE2	Cisco Catalyst C9300-24T-A	Branch Core / Distribution Layer 3 — Secondary	10.20.70.3/24	Cisco IOS XE 17.x	TEST-CENTER / Transparent
BR-GF-SW1	Cisco Catalyst WS-C2960-24PC-L	Branch Ground Floor Access Switch	10.20.70.11/24	Cisco IOS 15.x	TEST-CENTER / Transparent
BR-F1-SW1	Cisco Catalyst WS-C2960-24PC-L	Branch Floor 1 Reception Access Switch	10.20.70.12/24	Cisco IOS 15.x	TEST-CENTER / Transparent
BR-GF-SW2	Cisco Catalyst WS-C2960-24PC-L	Branch Access / Support Switch	10.20.70.13/24	Cisco IOS 15.x	TEST-CENTER / Transparent
BR-F2-SW1	Cisco Catalyst WS-C2960-48PST-L	Floor 2 Examination / PC Access Switch	10.20.70.14/24	Cisco IOS 15.x	TEST-CENTER / Transparent
BR-F2-SW2	Cisco Catalyst WS-C2960-48PST-L	Floor 2 CCTV / PC Access Switch	10.20.70.15/24	Cisco IOS 15.x	TEST-CENTER / Transparent
BR-F2-SW3	Cisco Catalyst WS-C2960-48PST-L	Floor 2 CCTV / AP Access Switch	10.20.70.16/24	Cisco IOS 15.x	TEST-CENTER / Transparent
BR-F3-SW1	Cisco Catalyst WS-C2960-48PST-L	Floor 3 Examination / PC Access Switch	10.20.70.17/24	Cisco IOS 15.x	TEST-CENTER / Transparent
BR-F3-SW2	Cisco Catalyst WS-C2960-48PST-L	Floor 3 CCTV / PC Access Switch	10.20.70.18/24	Cisco IOS 15.x	TEST-CENTER / Transparent
BR-F3-SW3	Cisco Catalyst WS-C2960-48PST-L	Floor 3 CCTV / AP Access Switch	10.20.70.19/24	Cisco IOS 15.x	TEST-CENTER / Transparent
BR-F4-SW1	Cisco Catalyst WS-C2960-48PST-L	Floor 4 IT / Administration Access Switch	10.20.70.20/24	Cisco IOS 15.x	TEST-CENTER / Transparent

HQ / Main Building – Core Switch Port Configuration

Port	Description	Access	VLAN	Trunk	EtherChannel	Native VLAN	Enabled
Gi1/0/1	Trunk Uplink → Access Switches	-	-	Yes	-	999	Yes
Gi1/0/2	Layer 3 Routed Link → ASA3	-	Routed	No	-	N/A	Yes
Gi1/0/3	Layer 3 EtherChannel Member → Peer Core Switch	-	Routed	No	Port-Channel1	N/A	Yes
Gi1/0/4	Layer 3 EtherChannel Member → Peer Core Switch	-	Routed	No	Port-Channel1	N/A	Yes

HQ / Main Building – Access Switch Configuration

Port	Description	Access	VLAN	Trunk	EtherChannel	Native VLAN	Enabled
Gi0/1	Trunk Uplink → HQ-CORE1	-	-	Yes	-	999	Yes
Gi0/2	Trunk Uplink → HQ-CORE2	-	-	Yes	-	999	Yes
Fa0/3–Fa0/24	Endpoint Connections → PCs / Cameras / Printers	Yes	According to Endpoint VLAN	No	-	N/A	Yes
Unused Ports	Unused / Reserved Ports	Yes	999	No	-	N/A	No — Shutdown

Branch 2 – Core Switch Port Configuration

Port	Description	Access	VLAN	Trunk	EtherChannel	Native VLAN	Enabled
Gi1/0/1	Trunk Uplink → Branch Access Switches	-	-	Yes	-	999	Yes
Gi1/0/2	Layer 3 Routed Link → ASA22	-	Routed	No	-	N/A	Yes
Gi1/0/3	Layer 3 EtherChannel Member → Peer Core Switch	-	Routed	No	Port-Channel1	N/A	Yes
Gi1/0/4	Layer 3 EtherChannel Member → Peer Core Switch	-	Routed	No	Port-Channel1	N/A	Yes
Gi1/0/5	Server Connection → Branch Server 1	-	60	No	-	N/A	Yes
Gi1/0/6	Server Connection → Branch Server 2	-	60	No	-	N/A	Yes

Branch 2 – Access Switch Configuration

Port	Description	Access	VLAN	Trunk	EtherChannel	Native VLAN	Enabled
Gi0/1	Trunk Uplink → BR-CORE1	-	-	Yes	-	999	Yes
Gi0/2	Trunk Uplink → BR-CORE2	-	-	Yes	-	999	Yes
Fa0/3–Fa0/48	Endpoint Connections → PCs / Cameras / Printers	Yes	According to Endpoint VLAN	No	-	N/A	Yes
Unused	Unused / Reserved Ports	Yes	999	No	-	N/A	No —

Port	Description	Access	VLAN	Trunk	EtherChannel	Native VLAN	Enabled
Ports							Shutdown

7.4 Firewall / ASA Device Documentation

Device	Model	Description	Location	Inside IP	Outside IP	DMZ IP	HA Status
ASA1	Cisco ASA — Simulator Model Not Supplied	HQ Security Edge / DMZ	HQ Ground Floor	10.10.255.2/30 → ASA3	10.10.252.1/30 → HQ-R1 10.10.252.5/30 → HQ-R2	DMZ interface → DMZ Server	Standalone / HA Not Supplied
ASA3	Cisco ASA — Simulator Model Not Supplied	HQ Security Edge / Core Boundary	HQ Ground Floor	10.10.254.5/30 → HQ-CORE1 10.10.254.2/30 → HQ-CORE2	10.10.255.1/30 → ASA1	Not Supplied	Standalone / HA Not Supplied
ASA22	Cisco ASA — Simulator Model Not Supplied	Branch Security Edge	Secondary Branch	Core-side interfaces → BR-CORE1 / BR-CORE2	WAN interfaces → BR-R1 / BR-R2	Not Supplied	Standalone / HA Not Supplied

7.5 End-System Documentation

Device	OS	Services	IPv4 / Prefix	IPv6 / Prefix	Default Gateway	DNS
Server0 / Core_Infrastructure_Server	Server OS — exact version not supplied	AD DS, DNS, DHCPv4 / DHCPv6 (HQ), NTP, NMS/Zabbix	10.10.60.10 /24	2001:db8:acad:60::10/64	10.10.60.1 / 2001:db8:acad:60::1	10.10.60.10 / 2001:db8:acad:60::10
Server1	Server OS — exact version not supplied	CCTV/NVR, Database, Syslog, SFTP	10.10.60.11 /24	2001:db8:acad:60::11/64	10.10.60.1 / 2001:db8:acad:60::1	10.10.60.10 / 2001:db8:acad:60::10
Server2	Server OS — exact version not supplied	Web Server, HTTPS, IP-PBX / FreePBX-Asterisk	10.10.60.12 /24	2001:db8:acad:60::12/64	10.10.60.1 / 2001:db8:acad:60::1	10.10.60.10 / 2001:db8:acad:60::10
Server3	Server OS — exact version not supplied	Backup Server	10.10.60.13 /24	2001:db8:acad:60::13/64	10.10.60.1 / 2001:db8:acad:60::1	10.10.60.10 / 2001:db8:acad:60::10

Device	OS	Services	IPv4 / Prefix	IPv6 / Prefix	Default Gateway	DNS
Camera_Server	Server OS — exact version not supplied	CCTV / NVR Service	10.10.60.14 /24	2001:db8:acad:60::14/64	10.10.60.1 / 2001:db8:acad:60::1	10.10.60.10 / 2001:db8:acad:60::10
Branch_DHCP_Server	Server OS — exact version not supplied	Local DHCPv4 /DHCPv6	10.20.60.10 /24	2001:db8:acad:160::10/64	10.20.60.1 / 2001:db8:acad:160::1	10.10.60.10 / 2001:db8:acad:60::10

7.6 Network Hardware, Roles, and Pricing

Category	Manufacturer	Series / Model	Type / Form Factor	Key Specifications	Qty	Unit Price	Estimated Total
Layer 3 Core / Distribution Switch — HQ	Cisco	Catalyst C9300-24T-A	Managed Layer 3, 1RU	24 × Gigabit Ethernet; SVI; OSPF; HSRP; ACL; EtherChannel; VLAN; STP; QoS	2	≈ USD 1,786	≈ USD 3,572
Layer 3 Core / Distribution Switch — Branch	Cisco	Catalyst C9300-24T-A	Managed Layer 3, 1RU	24 × Gigabit Ethernet; SVI; OSPF; HSRP; ACL; EtherChannel; VLAN; STP; QoS	2	≈ USD 1,786	≈ USD 3,572
Firewall — HQ Internet Edge	Cisco	Firepower 1000 / FPR1010-NGFW-K9	NGFW	Cisco FTD; Internet Edge; NAT; VPN; ACL; Threat Defense	1	USD 970–1,080	USD 970–1,080
Firewall — HQ DMZ/LAN Boundary	Cisco	Firepower 1000 / FPR1120-NGFW-K9	NGFW	Cisco FTD; DMZ–LAN boundary; inspection; VPN; NAT	1	USD 2,300–2,550	USD 2,300–2,550
Firewall — Secondary Branch	Cisco	Firepower 1000 / FPR1010-NGFW-K9	NGFW	Branch Internet Edge; NAT; VPN; ACL; Threat Defense	1	USD 970–1,080	USD 970–1,080
WAN Router — HQ	Cisco	Catalyst 8200 / C8200-1N-4T	Enterprise Edge Router, 1RU	WAN; IPsec VPN; SD-WAN; OSPF/BGP; Failover	2	≈ USD 2,400	≈ USD 4,800
WAN Router — Secondary Branch	Cisco	Catalyst 8200 / C8200-1N-4T	Enterprise Edge Router, 1RU	WAN; IPsec VPN; SD-WAN; OSPF/BGP; Failover	2	≈ USD 2,400	≈ USD 4,800
Access Switch — 48-Port PoE — HQ	Cisco	Catalyst 2960 / WS-C2960-48PST-L	Managed Layer 2, 1U	48 × Fast Ethernet; PoE; Gigabit uplink; VLAN/STP/QoS/Port Security	10	≈ USD 300	≈ USD 3,000

Category	Manufacturer	Series / Model	Type / Form Factor	Key Specifications	Qty	Unit Price	Estimated Total
Access Switch — 48-Port PoE — Branch	Cisco	Catalyst 2960 / WS-C2960-48PST-L	Managed Layer 2, 1U	48 × Fast Ethernet; PoE; Gigabit uplink; VLAN/STP/QoS/Port Security	7	≈ USD 300	≈ USD 2,100
Access Switch — 24-Port PoE — HQ	Cisco	Catalyst 2960 / WS-C2960-24PC-L	Managed Layer 2, 1U	24 × Fast Ethernet; PoE; Gigabit uplink; VLAN/STP/QoS/Port Security	3	≈ USD 200	≈ USD 600
Access Switch — 24-Port PoE — Branch	Cisco	Catalyst 2960 / WS-C2960-24PC-L	Managed Layer 2, 1U	24 × Fast Ethernet; PoE; Gigabit uplink; VLAN/STP/QoS/Port Security	3	≈ USD 200	≈ USD 600
Server0 — Core Infrastructure — HQ	Cisco	UCS C220 M7 SFF / UCSC-C220-M7S	1RU Rack Server	AD DS; DNS; DHCP; NTP; NMS	1	≈ USD 2,000	≈ USD 2,000
Server1 — Storage and Monitoring — HQ	Cisco	UCS C220 M7 SFF / UCSC-C220-M7S	1RU Rack Server	CCTV/NVR; Database; Syslog; SFTP; High I/O Storage	1	USD 8,000–12,000	USD 8,000–12,000
Server2 — Applications — HQ	Cisco	UCS C220 M7 SFF / UCSC-C220-M7S	1RU Rack Server	Web Server; HTTPS; IP-PBX	1	≈ USD 1,500	≈ USD 1,500
Server3 — Backup — HQ	Cisco	UCS C220 M6 SFF / UCSC-C220-M6S	1RU Rack Server	Backup Server; RAID; ≈15TB Storage	1	USD 3,500–5,000	USD 3,500–5,000
Branch DHCP Server	Cisco	UCS C220 M7 SFF / UCSC-C220-M7S	1RU Rack Server	Local DHCP Service for Secondary Branch	1	≈ USD 3,575–4,500	≈ USD 3,575–4,500
Branch Camera Server	Cisco	UCS C220 M7 SFF / UCSC-C220-M7S	1RU Rack Server	Local CCTV/NVR and Camera Recording	1	≈ USD 5,000–7,000	≈ USD 5,000–7,000
TOTAL PROJECT HARDWARE	—	—	—	—	39	—	≈ USD 50,259–59,154

Device / Role	Logical Use	Physical Placement	Packet Tracer Mapping	Status / Note
FPR1010-NGFW-K9 — HQ	Internet Edge Security	HQ Ground Floor — Firewalls Rack	ASA1	Hardware specification proposed; ASA1 is the logical Packet Tracer representation

Device / Role	Logical Use	Physical Placement	Packet Tracer Mapping	Status / Note
FPR1120-NGFW-K9 — HQ	DMZ ↔ LAN Security Boundary	HQ Ground Floor — Firewalls Rack	ASA3	Hardware specification proposed; ASA3 represents the HQ DMZ/LAN security boundary
FPR1010-NGFW-K9 — Branch	Branch Security Edge	Secondary Branch — Security Rack	ASA22	Hardware specification proposed; ASA22 is the logical Packet Tracer representation
C8200-1N-4T — HQ	WAN Routing, Internet Connectivity, IPsec VPN, SD-WAN	HQ Router Rack	HQ-R1, HQ-R2	Quantity: 2 routers at HQ
C8200-1N-4T — Branch	WAN Routing, Internet Connectivity, IPsec VPN, SD-WAN	Secondary Branch Router Rack	BR-R1, BR-R2	Quantity: 2 routers at the secondary branch
C9300-24T-A — HQ	Layer 3 Core / Distribution, SVI, OSPF, HSRP, ACL and EtherChannel	HQ Ground Floor — Core Rack	HQ-CORE1, HQ-CORE2	Quantity: 2 Layer 3 switches at HQ
C9300-24T-A — Branch	Layer 3 Core / Distribution, SVI, OSPF, HSRP, ACL and EtherChannel	Secondary Branch — Core Rack	BR-CORE1, BR-CORE2	Quantity: 2 Layer 3 switches at the secondary branch
WS-C2960-48PST-L — HQ	Access Layer for PCs, CCTV cameras, APs and PoE devices	HQ floor-level access racks	HQ-F2-SW1–SW3, HQ-F3-SW1–SW3, HQ-F4-SW1–SW3, HQ-F5-SW1	Quantity: 10 switches at HQ
WS-C2960-48PST-L — Branch	Access Layer for PCs, CCTV cameras, APs and PoE devices	Secondary Branch access racks	BR-F2-SW1–SW3, BR-F3-SW1–SW3, BR-F4-SW1	Quantity: 7 switches at the secondary branch
WS-C2960-24PC-L — HQ	Access Layer for smaller HQ areas and lower-density endpoints	HQ floor-level access racks	HQ-GF-SW1, HQ-GF-SW2, HQ-F1-SW1	Quantity: 3 switches at HQ
WS-C2960-24PC-L — Branch	Access Layer for smaller branch areas and lower-density endpoints	Secondary Branch access racks	BR-GF-SW1, BR-GF-SW2, BR-F1-SW1	Quantity: 3 switches at the secondary branch

7.7 Server Hardware, Services, and Capacity

Section	Server / Cost Item	Role / Service	Model / PID or Calculation	CPU / Virtual CPU	RAM	Physical / Virtual Storage	Network Placement	Form Factor / Qty
Physical Hardware	Server0	Core Infrastructure	Cisco UCS C220 M7 SFF / UCSC-C220-M7S	1 × Intel Xeon Gold 6544Y; 16 Core; 3.6 GHz	32 GB DDR5 ECC	480 GB SSD	HQ Data Center / VLAN 60 — Servers	1U / Qty 1
Physical Hardware	Server1	CCTV/NVR, Database, Syslog, SFTP	Cisco UCS C220 M7 SFF / UCSC-C220-M7S	1 × Intel Xeon Gold 6544Y; 16 Core; 3.6 GHz	16 GB DDR5 ECC	256 GB SSD	HQ Data Center / VLAN 60 — Servers	1U / Qty 1
Physical Hardware	Server2	Web Server, HTTPS, IP-PBX / FreePBX-Asterisk	Cisco UCS C220 M7 SFF / UCSC-C220-M7S	2 × Intel Xeon Gold; 24 Core each; 48 Core total	128 GB DDR5 ECC	10 × 2.5-inch NVMe SSD; up to 15.36 TB	HQ Data Center / VLAN 60 — Servers	1U / Qty 1
Physical Hardware	Server3	Backup Server	Cisco UCS C220 M6 SFF / UCSC-C220-M6S	1 × Intel Xeon Silver 4309Y; 8 Core	16 GB DDR4 ECC	Approximately 15 TB SAS HDD; Cisco 12G SAS RAID	HQ Data Center / VLAN 60 — Servers	1U / Qty 1
Virtual Services	Server0	AD DS; DNS; DHCPv4/DHCPv6; NTP; Zabbix NMS	Hosted on Server0	12 vCPU	32 GB allocated	450 GB NVMe allocated	Core Infrastructure / Server VLAN	Virtual workload
Virtual Services	Server2	Web Server; HTTPS; IP-PBX / FreePBX-Asterisk	Hosted on Server2	8 vCPU	16 GB allocated	256 GB NVMe allocated	HQ Data Center / VLAN 60 — Servers	Virtual workload
Virtual	Server2	Database —	Hosted	6 vCPU	32 GB	2 TB NVMe	Server	Virtual

Section	Server / Cost Item	Role / Service	Model / PID or Calculation	CPU / Virtual CPU	RAM	Physical / Virtual Storage	Network Placement	Form Factor / Qty
Services		MySQL/PostgreSQL	on Server2		allocated	allocated	VLAN	workload
Virtual Services	Server1	CCTV/NVR — Milestone/Shinobi	Hosted on Server1	8 vCPU	32 GB allocated	10 TB NVMe allocated	Camera Services / Server VLAN	Virtual workload
Virtual Services	Server1	SFTP — OpenSSH	Hosted on Server1	2 vCPU	8 GB allocated	2 TB NVMe allocated	Server VLAN	Virtual workload
Virtual Services	Server1	Syslog — Rsyslog/ELK	Hosted on Server1	4 vCPU	16 GB allocated	1 TB NVMe allocated	Management / Server VLAN	Virtual workload
Virtual Services	Server3	Backup Server	Hosted on Server3	6 vCPU	16 GB allocated	15+ TB SATA/SAS allocated	HQ Data Center / VLAN 60 — Servers	Virtual workload
Branch Servers	Branch_DHCP_Server	Local DHCPv4/DHCPv6 Service	Model and PID not supplied	Not supplied	Not supplied	Not supplied	Secondary Branch / VLAN 60 — Servers	Qty 1
Branch Servers	Camera_Server	Local CCTV/NVR Service	Model and PID not supplied	Not supplied	Not supplied	Not supplied	Secondary Branch / VLAN 60 — Servers	Qty 1

7.8 Cabling and Connectivity Documentation

Location / Link Group	Connected Equipment	Cable / Link Type	Port / Count Record	Design Record / Status	Technical Fields Assigned in the final implementation
General — Endpoint Connections	PCs, IP Cameras, Printers, WAPs	Cat6 UTP	According to the number of endpoint points	Specified in the physical design	Exact cable length, certification result, outlet ID, patch-panel port, switch port
General — Rack / Outlet Patching	Access Switch ↔ Patch Panel ↔ Outlet / Endpoint	Cat6 Patch Cable / Patch Panel Cabling	According to the Patch Panel ports	Generally specified in the rack and floor design	Patch-cord length, labeling scheme, patch-panel numbering, outlet numbering
General — Backbone / Uplinks	Access Switch ↔ Core/Distribution; Core ↔ Security	Fiber Optic	According to the number of uplink	Specified in the physical design	Single-mode / Multimode, fiber count, connector

Location / Link Group	Connected Equipment	Cable / Link Type	Port / Count Record	Design Record / Status	Technical Fields Assigned in the final implementation
	Devices; Inter-rack links		connections		type, SFP/optic type, link distance
General — Point-to-Point	Router ↔ Firewall; Firewall ↔ Core	Copper Ethernet / RJ-45	According to the interfaces used	Must be verified against the physical design and implementation	Cable category, length, connector, port labeling, negotiated speed
HQ Ground Floor — Control Unit	HQ-GF-SW1 ↔ Examination PCs	Cat6 UTP	5 PC ports	Design specified	Exact outlet, patch-panel and switch-port mapping
HQ Ground Floor — Control Unit	HQ-GF-SW1 ↔ CCTV Cameras	Cat6 UTP	8 camera ports	Design specified	Exact camera, outlet and switch-port mapping
HQ Ground Floor — Control Unit	HQ-GF-SW1 ↔ Printer	Cat6 UTP	1 printer port	Design specified	Exact outlet, patch-panel and switch-port mapping
HQ Ground Floor — Control Unit Uplinks	HQ-GF-SW1 ↔ Core/Distribution	Fiber Optic	2 uplink ports	Design specified	Exact switch ports, fiber type and patch-panel mapping
HQ Ground Floor — Server Area	HQ-GF-SW2 ↔ Servers	Cat6 UTP	5 server ports	Design specified	Exact server and switch-port mapping
HQ Ground Floor — Server Area Uplinks	HQ-GF-SW2 ↔ Core/Distribution	Fiber Optic	2 uplink ports	Design specified	Exact switch ports and fiber details
HQ Ground Floor — Core to Access Layer	HQ-CORE1 ↔ HQ Access Switches	Fiber Optic	13 access connections	Design specified; exact port mapping assigned	Exact core/access ports, fiber type and patch-panel details
HQ Ground Floor — Core to Access Layer	HQ-CORE2 ↔ HQ Access Switches	Fiber Optic	13 access connections	Design specified; exact port mapping assigned	Exact core/access ports, fiber type and patch-panel details
HQ Ground Floor — Core Interconnect	HQ-CORE1 ↔ HQ-CORE2	Fiber Optic	2 inter-core member ports per Layer 3 switch	Design specified; EtherChannel documented separately	Exact fiber/optic details and patch-panel mapping
HQ Ground Floor — Core to Security	HQ-CORE1 / HQ-CORE2 ↔ ASA3	Fiber Optic	1 security-edge link from each Layer 3 switch	Design/configuration documented	Exact physical media and patch-panel details
HQ Ground Floor — WAN Security Edge	HQ-R1 / HQ-R2 ↔ ASA1	Ethernet	2 router-to-firewall links	Design specified	Exact cable type, interface and patch-panel records
HQ Floor 1 — Reception	HQ-F1-SW1 ↔ PCs	Cat6 UTP	5 PC ports	Design specified	Exact outlet, patch-panel and switch-

Location / Link Group	Connected Equipment	Cable / Link Type	Port / Count Record	Design Record / Status	Technical Fields Assigned in the final implementation
					port mapping
HQ Floor 1 — Reception	HQ-F1-SW1 ↔ CCTV Cameras	Cat6 UTP	5 camera ports	Design specified	Exact camera, outlet and switch-port mapping
HQ Floor 1 — Reception	HQ-F1-SW1 ↔ Printers	Cat6 UTP	5 printer ports	Design specified	Exact outlet, patch-panel and switch-port mapping
HQ Floor 1 — Reception Uplink	HQ-F1-SW1 ↔ Core/Distribution	Fiber Optic	2 uplink ports	Design specified	Exact core ports and fiber details
HQ Floors 2–4 — Examination Areas	Access Switches ↔ Examination PCs	Cat6 UTP	46 PC ports per floor	Design specified	Exact endpoint, patch-panel and switch-port mapping
HQ Floors 2–4 — Examination Areas	Access Switches ↔ PC/CCTV Endpoints	Cat6 UTP	46 endpoint ports per floor — 8 PCs + 38 Cameras	Design specified	Exact endpoint and switch-port mapping
HQ Floors 2–4 — Examination Areas	Access Switches ↔ CCTV Cameras	Cat6 UTP	22 camera ports per floor	Design specified	Exact camera and switch-port mapping
HQ Floors 2–4 — Examination Areas	Access Switches ↔ Wireless Access Points	Cat6 UTP	6 WAP ports per floor	Design specified	Exact WAP and switch-port mapping
HQ Floors 2–4 — Access Uplinks	Access Switches ↔ Core/Distribution	Fiber Optic	2 uplink ports per access switch	Design specified	Exact switch-to-core mapping, fiber type and optic details
HQ Floor 5 — IT / Administration	HQ-F5-SW1 ↔ PCs	Cat6 UTP	19 PC ports	Design specified	Exact outlet, patch-panel and switch-port mapping
HQ Floor 5 — IT / Administration	HQ-F5-SW1 ↔ CCTV Cameras	Cat6 UTP	10 camera ports	Design specified	Exact camera and switch-port mapping
HQ Floor 5 — IT / Administration	HQ-F5-SW1 ↔ Printers	Cat6 UTP	3 printer ports	Design specified	Exact outlet, patch-panel and switch-port mapping
HQ Floor 5 — IT / Administration Uplink	HQ-F5-SW1 ↔ Core/Distribution	Fiber Optic	2 uplink ports	Design specified	Exact core ports and fiber details
Branch Ground Floor — Access	BR-GF-SW1 ↔ PCs	Cat6 UTP	5 PC ports	Design specified	Exact outlet, patch-panel and switch-port mapping
Branch Ground Floor — Access	BR-GF-SW1 ↔ CCTV Cameras	Cat6 UTP	8 camera ports	Design specified	Exact camera, outlet and switch-port mapping

Location / Link Group	Connected Equipment	Cable / Link Type	Port / Count Record	Design Record / Status	Technical Fields Assigned in the final implementation
Branch Ground Floor — Access	BR-GF-SW1 ↔ Printer	Cat6 UTP	1 printer port	Design specified	Exact outlet, patch-panel and switch-port mapping
Branch Ground Floor — Access Uplinks	BR-GF-SW1 ↔ BR-CORE1 / BR-CORE2	Fiber Optic	2 uplink ports	Design specified	Exact core ports and fiber details
Branch Floor 1 — Reception	BR-F1-SW1 ↔ PCs	Cat6 UTP	5 PC ports	Design specified	Exact outlet, patch-panel and switch-port mapping
Branch Floor 1 — Reception	BR-F1-SW1 ↔ CCTV Cameras	Cat6 UTP	5 camera ports	Design specified	Exact camera, outlet and switch-port mapping
Branch Floor 1 — Reception	BR-F1-SW1 ↔ Printers	Cat6 UTP	5 printer ports	Design specified	Exact outlet, patch-panel and switch-port mapping
Branch Floor 1 — Reception Uplink	BR-F1-SW1 ↔ BR-CORE1 / BR-CORE2	Fiber Optic	2 uplink ports	Design specified	Exact core ports and fiber details
Branch Floors 2–3 — Examination Areas	Access Switches ↔ Examination PCs	Cat6 UTP	46 PC ports per floor	Design specified	Exact endpoint, patch-panel and switch-port mapping
Branch Floors 2–3 — Examination Areas	Access Switches ↔ PC/CCTV Endpoints	Cat6 UTP	46 endpoint ports per floor — 8 PCs + 38 Cameras	Design specified	Exact endpoint and switch-port mapping
Branch Floors 2–3 — Examination Areas	Access Switches ↔ CCTV Cameras	Cat6 UTP	22 camera ports per floor	Design specified	Exact camera and switch-port mapping
Branch Floors 2–3 — Examination Areas	Access Switches ↔ Wireless Access Points	Cat6 UTP	6 WAP ports per floor	Design specified	Exact WAP and switch-port mapping
Branch Floors 2–3 — Access Uplinks	Access Switches ↔ BR-CORE1 / BR-CORE2	Fiber Optic	2 uplink ports per access switch	Design specified	Exact switch-to-core mapping, fiber type and optic details
Branch Floor 4 — IT / Administration	BR-F4-SW1 ↔ PCs	Cat6 UTP	19 PC ports	Design specified	Exact outlet, patch-panel and switch-port mapping
Branch Floor 4 — IT / Administration	BR-F4-SW1 ↔ CCTV Cameras	Cat6 UTP	10 camera ports	Design specified	Exact camera and switch-port mapping

Location / Link Group	Connected Equipment	Cable / Link Type	Port / Count Record	Design Record / Status	Technical Fields Assigned in the final implementation
Branch Floor 4 — IT / Administration	BR-F4-SW1 ↔ Printers	Cat6 UTP	3 printer ports	Design specified	Exact outlet, patch-panel and switch-port mapping
Branch Floor 4 — IT / Administration Uplink	BR-F4-SW1 ↔ BR-CORE1 / BR-CORE2	Fiber Optic	2 uplink ports	Design specified	Exact core ports and fiber details
Branch Ground Floor — Core to Security	BR-CORE1 / BR-CORE2 ↔ ASA22	Fiber Optic	1 security-edge link from each Layer 3 switch	Design/configuration documented	Exact physical media and patch-panel details
Branch Ground Floor — WAN Security Edge	BR-R1 / BR-R2 ↔ ASA22	Ethernet	2 router-to-firewall links	Design specified	Exact cable type, interface and patch-panel records
Branch Ground Floor — Core Interconnect	BR-CORE1 ↔ BR-CORE2	Fiber Optic	2 inter-core member ports per Layer 3 switch	Design specified; EtherChannel documented separately	Exact fiber/optic details and patch-panel mapping
Branch Ground Floor — Core to Servers	BR-CORE1 / BR-CORE2 ↔ Branch Servers	Cat6 UTP	2 server connections	Design specified	Exact server, core-port and patch-panel mapping

7.9 Endpoint Device, VLAN, and Addressing Plan

Site	Floor / Area	Endpoint Type	Quantity	VLAN	IPv4 Network	IPv6 Prefix	Address Assignment
HQ	Floor 1 – Reception	PCs	5	10 – ADMIN	10.10.10.0/24	2001:db8:acad:10::/64	DHCPv4 + DHCPv6
HQ	Floors 2–4 – Examination	PCs	54 per floor	30 – TEST_CENTER	10.10.30.0/24	2001:db8:acad:30::/64	DHCPv4 + DHCPv6
HQ	Floor 5 – Administration	PCs	3	10 – ADMIN	10.10.10.0/24	2001:db8:acad:10::/64	DHCPv4 + DHCPv6
HQ	Floor 5 – IT / Employees	PCs	16	20 – EMPLOYEES	10.10.20.0/24	2001:db8:acad:20::/64	DHCPv4 + DHCPv6
HQ	All Floors	IP Cameras	203	40 – CCTV	10.10.40.0/24	2001:db8:acad:40::/64	DHCPv4 Reservation + DHCPv6
HQ	Floors 1 & 5	Printers	9	90 – PRINTERS_IOT	10.10.90.0/24	2001:db8:acad:90::/64	DHCPv4 Reservation + DHCPv6
Branch	Floor 1 –	PCs	5	10 – ADMIN	10.20.10.0/24	2001:db8:acad:110::/64	DHCPv4 +

Site	Floor / Area	Endpoint Type	Quantity	VLAN	IPv4 Network	IPv6 Prefix	Address Assignment
2	Reception						DHCPv6
Branch 2	Floors 2–3 – Examination	PCs	54 per floor	30 – TEST_CENTER	10.20.30.0/24	2001:db8:acad:130::/64	DHCPv4 + DHCPv6
Branch 2	Floor 4 – Administration	PCs	3	10 – ADMIN	10.20.10.0/24	2001:db8:acad:110::/64	DHCPv4 + DHCPv6
Branch 2	Floor 4 – IT / Employees	PCs	16	20 – EMPLOYEES	10.20.20.0/24	2001:db8:acad:120::/64	DHCPv4 + DHCPv6
Branch 2	All Floors	IP Cameras	143	40 – CCTV	10.20.40.0/24	2001:db8:acad:140::/64	DHCPv4 Reservation + DHCPv6
Branch 2	Floors 1 & 4	Printers	9	90 – PRINTERS_IOT	10.20.90.0/24	2001:db8:acad:190::/64	DHCPv4 Reservation + DHCPv6

7.10 — Representative Endpoint Addressing

Device ID	Site / Floor	Endpoint Type	VLAN	IPv4 Address	IPv6 Address	Addressing Method	Gateway
HQ-F1-PC-01	HQ / Floor 1 Reception	PC	10 – ADMIN	DHCPv4	DHCPv6	DHCPv4 + DHCPv6	10.10.10.1 / 2001:db8:acad:10::1
HQ-F2-PC-01	HQ / Floor 2 Examination	PC	30 – TEST_CENTER	DHCPv4	DHCPv6	DHCPv4 + DHCPv6	10.10.30.1 / 2001:db8:acad:30::1
HQ-F5-PC-01	HQ / Floor 5 IT	PC	20 – EMPLOYEES	DHCPv4	DHCPv6	DHCPv4 + DHCPv6	10.10.20.1 / 2001:db8:acad:20::1
HQ-F5-PC-17	HQ / Floor 5 Administration	PC	10 – ADMIN	DHCPv4	DHCPv6	DHCPv4 + DHCPv6	10.10.10.1 / 2001:db8:acad:10::1
HQ-F2-CAM-01	HQ / Floor 2	IP Camera	40 – CCTV	DHCPv4 Reservation	DHCPv6	DHCPv4 Reservation + DHCPv6	10.10.40.1 / 2001:db8:acad:40::1
HQ-F1-PRN-01	HQ / Floor 1 Reception	Printer	90 – PRINTERS_IOT	DHCPv4 Reservation	DHCPv6	DHCPv4 Reservation + DHCPv6	10.10.90.1 / 2001:db8:acad:90::1
BR-F1-PC-01	Branch 2 / Floor 1 Reception	PC	10 – ADMIN	DHCPv4	DHCPv6	DHCPv4 + DHCPv6	10.20.10.1 / 2001:db8:acad:110::1
BR-F2-	Branch 2 / Floor	PC	30 –	DHCPv4	DHCPv6	DHCPv4 +	10.20.30.1 /

Device ID	Site / Floor	Endpoint Type	VLAN	IPv4 Address	IPv6 Address	Addressing Method	Gateway
PC-01	2 Examination		TEST_CENTER			DHCPv6	2001:db8:acad:130::1
BR-F4-PC-01	Branch 2 / Floor 4 IT	PC	20 – EMPLOYEES	DHCPv4	DHCPv6	DHCPv4 + DHCPv6	10.20.20.1 / 2001:db8:acad:120::1
BR-F4-PC-17	Branch 2 / Floor 4 Administration	PC	10 – ADMIN	DHCPv4	DHCPv6	DHCPv4 + DHCPv6	10.20.10.1 / 2001:db8:acad:110::1
BR-F2-CAM-01	Branch 2 / Floor 2	IP Camera	40 – CCTV	DHCPv4 Reservation	DHCPv6	DHCPv4 Reservation + DHCPv6	10.20.40.1 / 2001:db8:acad:140::1
BR-F1-PRN-01	Branch 2 / Floor 1 Reception	Printer	90 – PRINTERS_IOT	DHCPv4 Reservation	DHCPv6	DHCPv4 Reservation + DHCPv6	10.20.90.1 / 2001:db8:acad:190::1

08. Routing and OSPFv2

Router	Router ID	Area	Neighbors	Passive Interfaces	MD5 Key ID	Cost / Bandwidth
BR-R1	3.3.3.3	0	BR-R2	LAN-facing interfaces	1	Cost 1 / GigabitEthernet
BR-R2	4.4.4.4	0	BR-R1	LAN-facing interfaces	1	Cost 1 / GigabitEthernet
HQ-R1	1.1.1.1	0	HQ-R2	LAN-facing interfaces	1	Cost 1 / GigabitEthernet
HQ-R2	2.2.2.2	0	HQ-R1	LAN-facing interfaces	1	Cost 1 / GigabitEthernet

09. ACL, NAT, and DMZ

9.1 ACL Policy

Rule ID	Source	Destination	Service	Action	Reason	Device / Interface	Direction
ACL-SSH-01	IT-ADMIN subnet	Network device VTY	SSH / TCP 22	Permit	Admin management only	All managed devices	Inbound
ACL-DMZ-01	Any / Internet	DMZ Web Server 10.10.100.10	HTTP / TCP 80	Permit	Public web access to DMZ server	ASA1 / outside-r1	Inbound
ACL-DMZ-02	Any / Internet	DMZ Web Server 10.10.100.10	HTTPS / TCP 443	Permit	Secure public web access to DMZ server	ASA1 / outside-r1	Inbound
ACL-DMZ-03	Any / Internet	DMZ Web Server 10.10.100.10	ICMP	Permit	Connectivity and reachability	ASA1 / outside-r1	Inbound
ACL-DMZ-04	EXAM-USERS	DMZ Web Server 10.10.100.10	HTTP / HTTPS	Permit	Allowed exam/support web access	ASA1	Inbound

Rule ID	Source	Destination	Service	Action	Reason	Device / Interface	Direction
ACL-DMZ-05	EXAM-USERS	Other DMZ services	Any other	Deny + Log	Protect DMZ services	ASA1	Inbound

9.3 NAT Policy

NAT ID	Type	Inside Local	Inside Global / Public	Purpose
NAT-01	PAT / NAT Overload	HQ Internal Networks	Internet	Translate
NAT-02	PAT / NAT Overload	Branch Internal Networks	Internet	Translate
NAT-03	Static NAT	DMZ Web Server 10.10.100.10	Public Web Address	1:1 Translation

10. WAN, IPsec VPN, and QoS

10.1 IPsec VPN

Tunnel ID	Endpoint A	Endpoint B	Protected Networks	Encryption / Auth	Failover
VPN-01	HQ-R1 — 10.10.250.2	BR-R1 — 10.20.250.2	HQ: 10.10.0.0/16 ↔ Branch: 10.20.0.0/16	IPsec: AES + SHA-HMAC / ISAKMP: AES, SHA, Pre- Shared Key, DH Group 2	ISP2: HQ 10.10.250.10 ↔ Branch 10.20.250.10 using VPN-MAP 20

10.2 QoS Policy

Class	Traffic Match	Marking	Queue / Policy	Interface	Expected Behavior
REALTIME	VoIP / IP-PBX and real-time voice traffic	DSCP EF	Strict Priority Queue (LLQ)	WAN	Lowest latency and jitter
WEB	HTTP / HTTPS	Default / DSCP BE	Default Queue	WAN	Best-effort delivery
MGMT	SSH / SNMP / Syslog	DSCP AF21	Guaranteed Bandwidth	WAN	Reliable network management traffic

11. Network Services

Service	Server / IP	Clients	Security	Evidence ID
NTP	Core_Infrastructure_Server — 10.10.60.10	All routers, switches, and firewalls	Restricted Management VLAN	NTP-01
Syslog	Server1 — 10.10.60.11	All network devices	Management ACL	SYSLOG-01
SNMPv3	Core_Infrastructure_Server — 10.10.60.10	Monitoring platform / Zabbix	AuthPriv	SNMP-01

12. Conclusion

The International Examination Center network is designed as a **structured, scalable, secure, and highly available enterprise network** connecting the Main Building and Branch through redundant WAN connectivity. The design uses dedicated VLANs and IP subnets to separate **administrative users, employees, examination systems, CCTV cameras, voice services, servers, network management, guest access, printers/IoT devices, and DMZ services**.

The network architecture integrates **multilayer switching, OSPFv2, Layer 3 EtherChannel, firewall security, ACLs, NAT/PAT, IPsec VPN, and QoS** to provide controlled communication, secure inter-site connectivity, traffic prioritization, and reliable network operation. Centralized services including **DHCPv4/DHCPv6, DNS, NTP, Syslog, and SNMPv3/Zabbix monitoring** support consistent management and monitoring across both sites.

The overall design provides a foundation that can support the current requirements of the International Examination Center while allowing additional users, devices, services, and network capacity to be incorporated as the organization grows.